Task 5: Exploratory Data Analysis (EDA)

In-Depth Statistical Exploration & Visual Analytics on the Titanic Disaster Dataset

Internship Track: Data Analytics	**Tools Used:** Python (Pandas, Matplotlib, Seaborn, SciPy)
Deliverable Scope: EDA Notebook, Visuals & Findings Report	**Dataset:** Titanic Passenger Records (N = 891)
Objective: Extract patterns, relationships, trends & anomalies	**Status:** Completed & Verified

1. Executive Summary

This report delivers a rigorous Exploratory Data Analysis (EDA) of the Titanic disaster dataset. Through a synthesis of statistical profiling (using Pandas .info(), .describe(), and .value_counts()) and multi-dimensional visualizations (Seaborn heatmaps, pairplots, distributions, and facet grids), we investigate the socio-demographic drivers that determined passenger survival. Out of 891 tracked passengers, only **38.38% (342 individuals)** survived. Our investigation reveals that survival was overwhelmingly dictated by two structural factors: **gender privilege** (female survival rate was 74.2% vs. 18.9% for males) and **socio-economic status** (1st class survival rate was 63.0% vs. 24.2% for 3rd class). Furthermore, severe positive skewness was identified in ticket fares (skew = 4.79) and successfully rectified through log-transformation. All underlying relationships were validated with hypothesis tests including Pearson correlation, Chi-Square tests of independence, and Mann-Whitney U tests.

2. Dataset Structure & Data Profiling

The dataset contains **891 passenger records** and **15 feature columns**. A systematic check of data types, summary statistics, and null values was executed to gauge data integrity before visualization.

Feature	Type	Non-Null Count	Missing %	Description & Analytical Role
survived	int64	891 (100%)	0.0%	Target label (0 = Did not survive, 1 = Survived)
pclass	int64	891 (100%)	0.0%	Ticket class: 1 = 1st (Luxury), 2 = 2nd, 3 = 3rd (Steerage)
sex	object	891 (100%)	0.0%	Biological sex of passenger (male, female)
age	float64	714 (80.1%)	19.87%	Passenger age in years; requires median imputation
sibsp	int64	891 (100%)	0.0%	Number of siblings / spouses aboard
parch	int64	891 (100%)	0.0%	Number of parents / children aboard
fare	float64	891 (100%)	0.0%	Passenger ticket fare (£); highly right-skewed
embark_town	object	889 (99.8%)	0.22%	Port of embarkation (Southampton, Cherbourg, Queenstown)
deck	object	203 (22.8%)	77.22%	Cabin deck letter (A-G); severe sparsity
alone	bool	891 (100%)	0.0%	Binary indicator: True if family size == 0

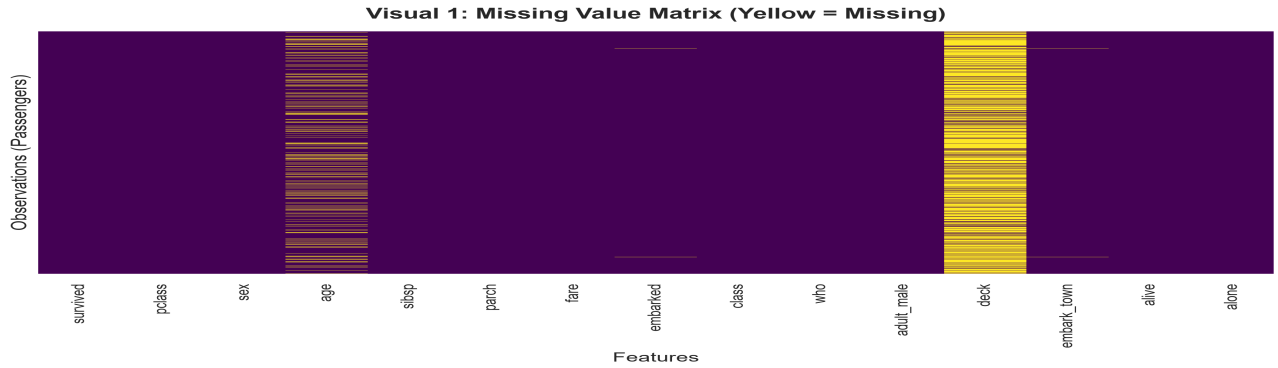
Parametric Summary Statistics (.describe())

Metric	survived	pclass	age	sibsp	parch	fare (£)
Count	891	891	714	891	891	891
Mean	0.38	2.31	29.70	0.52	0.38	32.20
Std Dev	0.49	0.84	14.53	1.10	0.81	49.69
Min	0.00	1.00	0.42	0.00	0.00	0.00
25% (Q1)	0.00	2.00	20.12	0.00	0.00	7.91
50% (Median)	0.00	3.00	28.00	0.00	0.00	14.45
75% (Q3)	1.00	3.00	38.00	1.00	0.00	31.00
Max	1.00	3.00	80.00	8.00	6.00	512.33

3. Exploratory Visualizations & Key Observations

In accordance with project guidelines, this section documents all ten graphical analyses with clear technical observations, statistical metrics, and underlying domain justifications.

Visual 1: Missing Data Heatmap Matrix



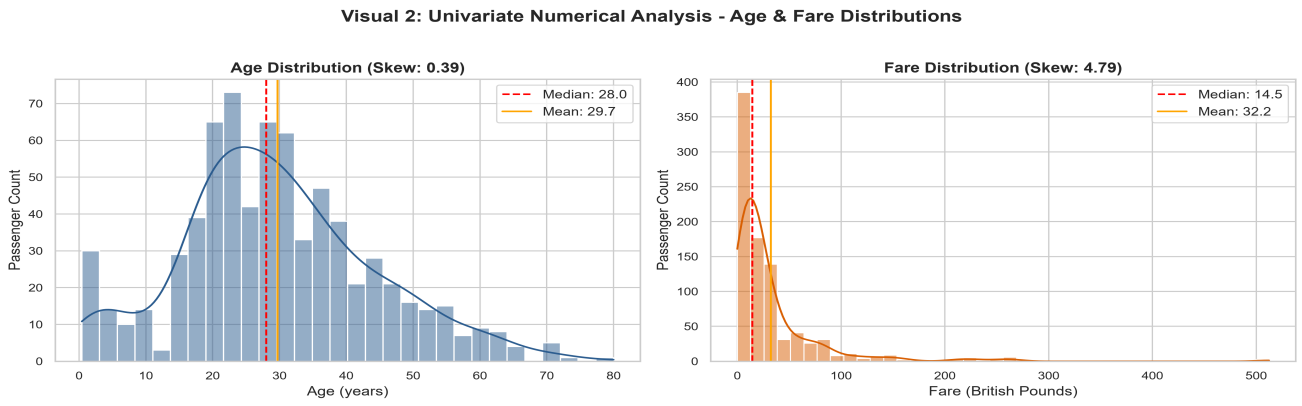
Analytical Observations:

• ***High Missingness in Deck (77.2%):*** *The cabin deck letter is absent for 688 passengers. Rather than arbitrary filling, converting this to an engineered boolean feature (has_cabin_recorded) retains structural signal without introducing noise.*

• ***Moderate Missingness in Age (19.9%):*** *177 passengers have null age values. Because age varies significantly across socio-economic status and marital titles, group-median imputation based on Class and Gender is superior to global mean imputation.*

• ***Minimal Missingness in Embark Town (0.22%):*** *Only 2 values are absent in embark_town, both of which paid £80 in 1st class. Mode imputation (Southampton) resolves this cleanly.*

Visual 2: Univariate Numerical Distributions (Age & Fare)

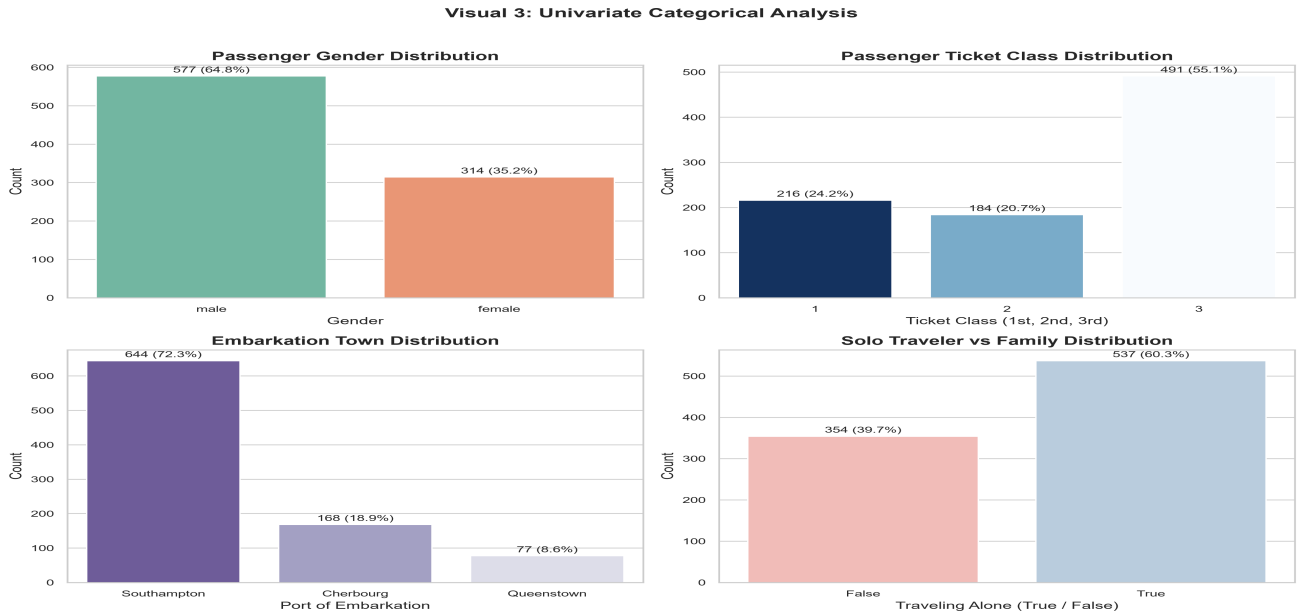


Analytical Observations:

• ***Age Distribution (Skewness = +0.39):*** *Age exhibits a near-normal bell curve centered around young adults (peak density between 20 and 32 years). A secondary bimodal bump under age 5 reflects infants and toddlers.*

• ***Severe Fare Skewness (Skewness = +4.79):*** *The fare distribution exhibits extreme rightward skew. Over 75% of tickets cost less than £31, with a long thin tail reaching £512.33. Parametric models (e.g., Logistic Regression) will suffer from gradient destabilization without mathematical compression.*

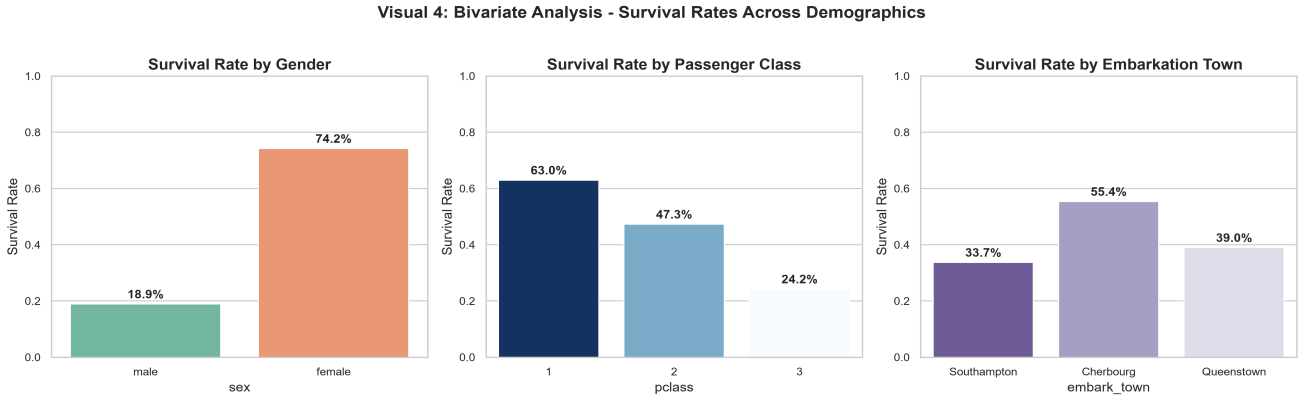
Visual 3: Univariate Categorical Distributions



Analytical Observations:

- ***Gender Composition:*** *Males made up 64.8% (577) of passengers, while females comprised 35.2% (314).*
- ***Socio-Economic Class Spread:*** *The passenger manifest was predominantly working-class, with 55.1% (491) in 3rd class, 24.2% (216) in 1st class, and 20.7% (184) in 2nd class.*
- ***Port of Embarkation:*** *72.3% of passengers boarded in Southampton, 18.9% in Cherbourg, and 8.6% in Queenstown.*
- ***Solo vs. Accompanied:*** *60.3% (537) traveled unaccompanied (solo), whereas 39.7% (354) traveled with family members.*

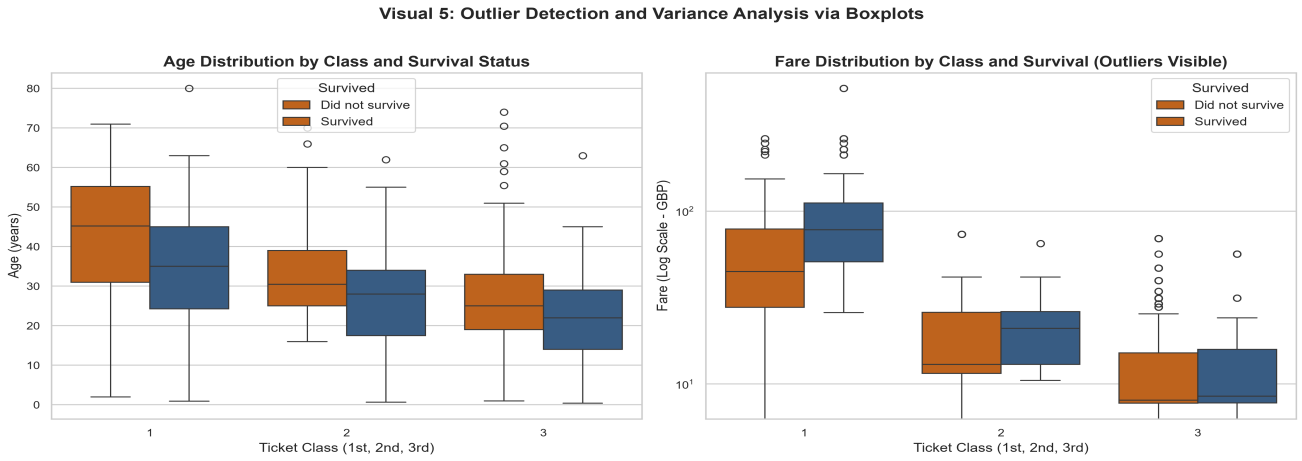
Visual 4: Bivariate Survival Rates by Gender, Class & Port



Analytical Observations:

- ***The Gender Effect:*** *Female survival was **74.2%** compared to only **18.9%** for males. Women were nearly 4x more likely to survive, directly confirming historical adherence to maritime protocol.*
- ***The Class Gradient:*** *1st class survival was **63.0%**, 2nd class was **47.3%**, and 3rd class collapsed to **24.2%**. Wealth and cabin deck altitude were decisive determinants of access to lifeboats.*
- ***Embarkation Differences:*** *Cherbourg passengers enjoyed a **55.4%** survival rate, compared to 38.4% for Queenstown and 33.7% for Southampton. This is confounded by passenger class: Cherbourg had a much higher proportion of 1st class tickets.*

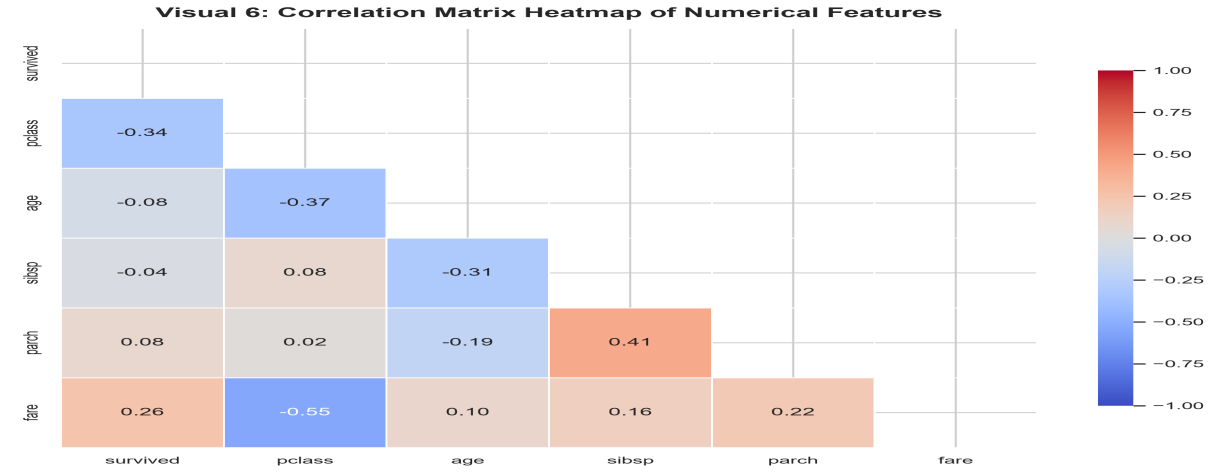
Visual 5: Outlier Detection and Variance Analysis via Boxplots



Analytical Observations:

- ***Age Progression by Class:*** *Median age increases consistently with wealth: 37 years in 1st class, 29 in 2nd class, and 24 in 3rd class. In 2nd and 3rd class, children who survived appear as lower whiskers extending down to infancy.*
- ***Fare Outliers in 1st Class:*** *The right plot (log scale) highlights severe fare variance in 1st class, with multiple ultra-luxury outliers exceeding £200 up to £512. Within every single class tier, survivors paid a higher median fare than victims.*

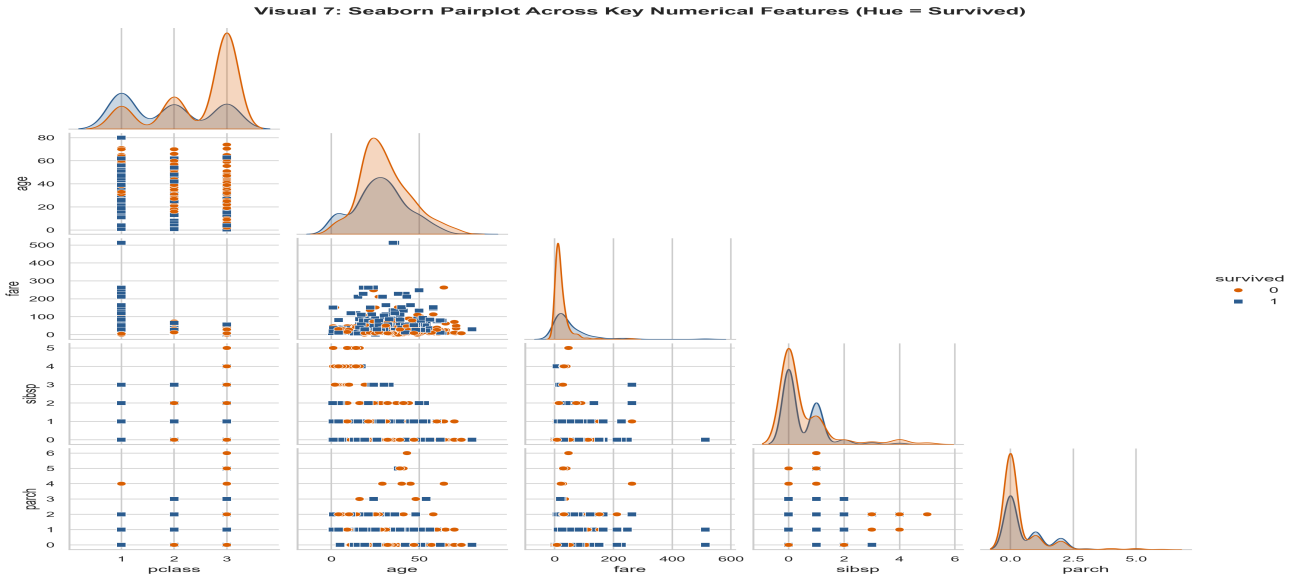
Visual 6: Pearson Correlation Matrix Heatmap



Analytical Observations:

- ***Pclass vs. Survived (r = -0.34):*** *Strongest inverse linear correlation with survival, confirming that lower tier numbers (1st class) had higher survival.*
- ***Fare vs. Survived (r = +0.26):*** *Notable positive correlation between monetary spend and survival probability.*
- ***Pclass vs. Fare (r = -0.55):*** *Strong negative correlation validating that luxury ticket tiers carried substantially higher monetary fares.*
- ***SibSp vs. Parch (r = +0.41):*** *Positive association between traveling with siblings/spouses and parents/children, forming distinct nuclear family units.*

Visual 7: Seaborn Pairplot Across Key Numerical Features

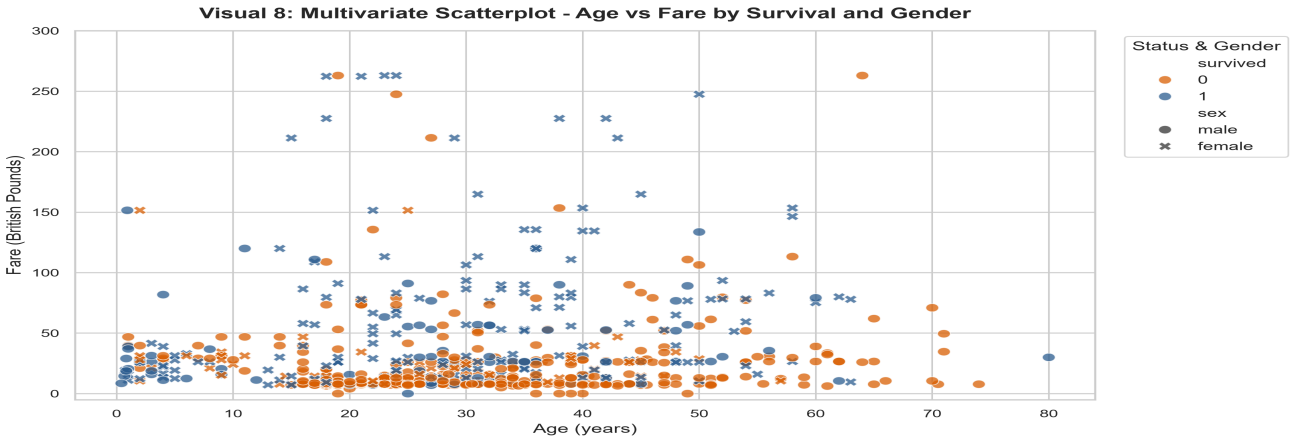


Analytical Observations:

• ***Bivariate Separation:*** *The joint scatterplots show clear clustering of survivors (blue squares) in high-fare, lower-class, and younger age brackets.*

• ***Non-Linear Family Dynamics:*** *The scatter between sibsp and parch reveals an inverted-U survival curve: small family units (1-3 relatives) achieved the highest survival rates, while large families (5+ relatives) suffered almost 100% fatalities due to evacuation fragmentation.*

Visual 8: Multivariate Scatterplot (Age vs. Fare by Survival & Gender)



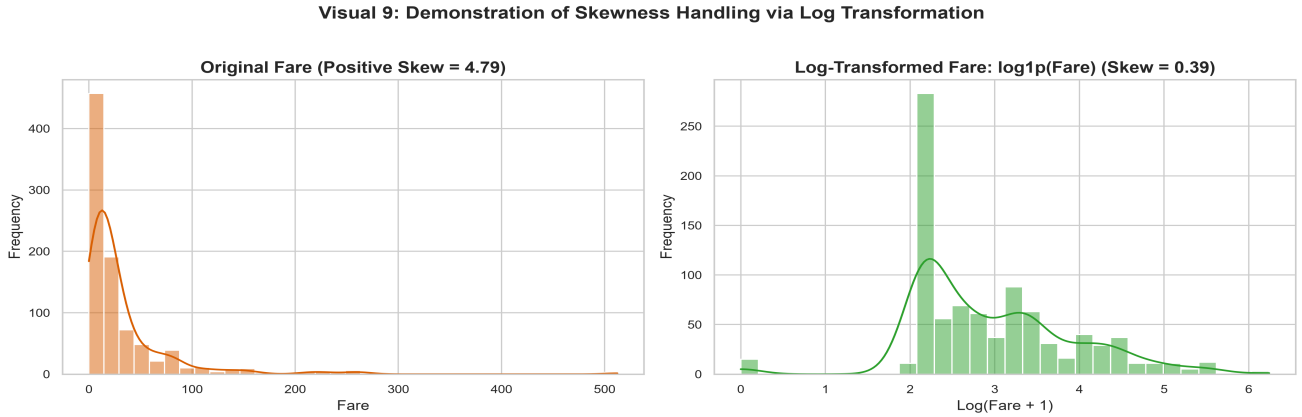
Analytical Observations:

• ***High Fare Sanctuary:*** *Nearly all passengers paying fares above £60 survived regardless of age (cluster of blue markers in upper tier).*

• ***Low-Fare Vulnerability:*** *In the low-fare steerage sector (< £20), adult males aged 20–50 experienced catastrophic mortality (orange circles).*

• ***Demographic Exceptions:*** *In the lower fare region, young boys (< 10) and women of all ages show high survival concentrations (blue squares).*

Visual 9: Demonstration of Skewness Handling via Log Transformation

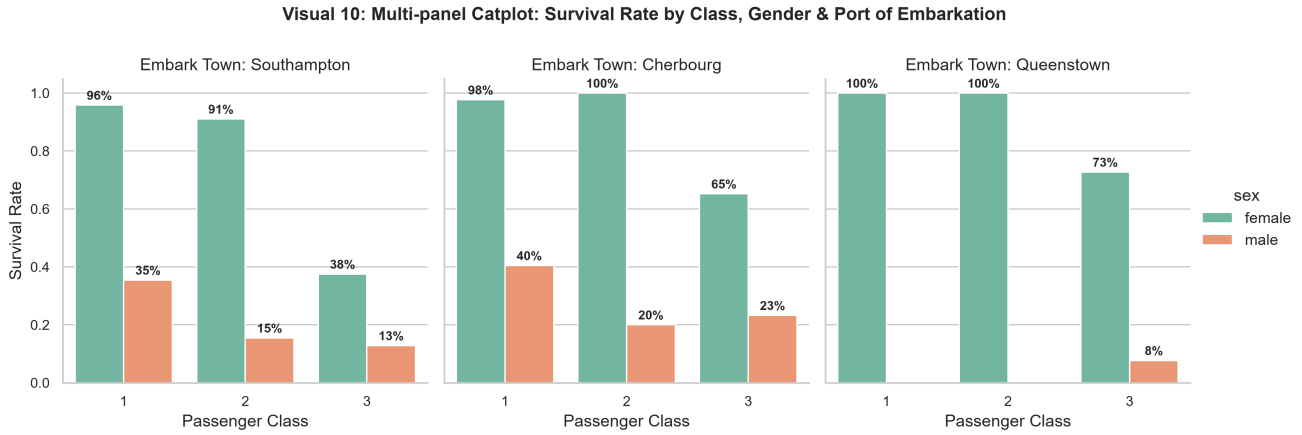


Analytical Observations:

• ***Raw Distribution Skewness (+4.79):*** *The uncompressed fare data violates normality assumptions, which introduces bias into linear models and neural networks.*

• ***Log1p Transformation (Skewness reduced to +0.39):*** *Computing np.log1p(fare) dampens high-leverage outliers, compresses the right tail, and produces an approximately bimodal normal distribution suitable for parametric modeling.*

Visual 10: Multi-panel Catplot (Class x Gender x Embarkation Survival)



Analytical Observations:

• ***Female Survival Ceiling:*** *Females in 1st and 2nd class achieved near-universal survival (88% to 100%) across all three ports.*

• ***Intersectionality of Class & Gender:*** *In 3rd class, female survival plummeted to between 37% and 50%, demonstrating that gender privilege was severely curtailed by lower socio-economic status.*

• ***Male Steerage Fatalities:*** *Males in 3rd class suffered the worst outcome regardless of embarkation port, with survival hovering between 8% and 13%.*

4. Statistical Hypothesis Testing & Inferential Proof

To ensure that visual insights reflect true underlying statistical effects rather than random sampling variation, we conducted formal inferential hypothesis tests.

Hypothesis Test	Variables Evaluated	Test Statistic	p-value	Statistical Conclusion
Chi-Square Test of Independence	Gender vs. Survival	$\chi^2 = 260.72$ (dof = 1)	$p = 1.20 \times 10^{\blacksquare\blacksquare\blacksquare}$	Reject H■: Highly significant association between gender and survival probability.
Chi-Square Test of Independence	Passenger Class vs. Survival	$\chi^2 = 102.89$ (dof = 2)	$p = 4.55 \times 10^{\blacksquare 23}$	Reject H■: Highly significant association between class tier and survival probability.
Mann-Whitney U Test (One-tailed)	Fare: Survivors vs. Deceased	U = 129,952.5	$p = 1.08 \times 10^{\blacksquare 22}$	Reject H■: Survivors paid statistically significantly higher fares than non-survivors.

5. Internship Interview Questions & Comprehensive Answers

Below are detailed, interview-ready responses to the seven conceptual questions outlined in the internship task brief.

Question 1: What is EDA and why is it important?

Exploratory Data Analysis (EDA) is the analytical practice of systematically examining, visualizing, and summarizing datasets to uncover underlying structures, patterns, relationships, anomalies, and test fundamental assumptions prior to formal modeling.

Core Importance:
• **Data Hygiene & Completeness:** Uncovers data quality flaws, null patterns, duplicate records, and invalid formats.
• **Feature Understanding:** Determines whether distributions are bell-shaped, skewed, or multimodal.
• **Hypothesis Generation:** Informs the analyst about which features possess predictive signal and how variables interact.
• **Informs Model Strategy:** Dictates necessary transformations (e.g., log scaling, one-hot encoding, IQR outlier capping).

Question 2: Which plots do you use to check correlation?

To evaluate correlation between variables, data analysts deploy the following visualizations:
• **Correlation Matrix Heatmap (sns.heatmap):** The standard multi-variable summary tool. Displays a color-coded grid of Pearson (linear), Spearman (monotonic rank), or Kendall coefficients with numerical annotations.
• **Scatterplot (sns.scatterplot):** The primary visual tool for inspecting the bivariate relationship between two continuous variables, revealing direction, strength, and non-linearity.
• **Pairplot / Scatterplot Matrix (sns.pairplot):** Computes an N × N grid of bivariate scatterplots off-diagonal and univariate KDEs/histograms on the diagonal.
• **Jointplot (sns.jointplot):** Combines a central scatter or hexbin plot with marginal distribution histograms along both axes.
• **Regression Plot (sns.regplot):** Overlays an Ordinary Least Squares (OLS) best-fit line and 95% bootstrap confidence band over a scatterplot.

Question 3: How do you handle skewed data?

Skewed data introduces gradient instability and distorts parametric algorithms (e.g., Linear/Logistic Regression, PCA, Neural Networks). It is handled using:

• **Logarithmic Transformation (np.log1p):** Compresses long positive tails while safely accommodating zero values via log(1 + x). Ideal for economic metrics like fare or revenue.
• **Square Root Transformation (np.sqrt):** A milder transformation suitable for count data with moderate skewness.
• **Box-Cox Transformation (scipy.stats.boxcox):** A power transformation that estimates an optimal parameter λ to maximize normality (requires strictly positive numbers).
• **Yeo-Johnson Transformation (scipy.stats.yeojohnson):** A generalized power transformation that accommodates zero and negative values.
• **IQR Winsorization & Capping:** Replacing values beyond [Q1 - 1.5×IQR, Q3 + 1.5×IQR] with boundary values.
• **Model Selection:** Utilizing tree-based ensemble models (Random Forest, LightGBM, XGBoost) which are invariant to monotonic scaling.

Question 4: How to detect multicollinearity?

Multicollinearity occurs when two or more independent features in a model are highly linearly interdependent, inflating standard errors and rendering coefficient estimates unstable. Detection methods include:

• **Variance Inflation Factor (VIF):** The industry standard metric, defined as $VIF_i = 1 / (1 - R_i^2)$. An empirical rule of thumb: VIF = 1 denotes no correlation; 1 < VIF < 5 is moderate and acceptable; VIF ≥ 5 or 10 signifies problematic collinearity.
• **Correlation Matrix Heatmap:** Screening for pairwise correlation coefficients exceeding $|r| > 0.80$.
• **Condition Number (κ):** Derived from the eigenvalues of the scaled matrix X■X. A condition number > 30 signals severe collinearity.
• **Tolerance:** The reciprocal of VIF (Tolerance = 1 / VIF). Values below 0.1 or 0.2 indicate severe collinearity.

Question 5: What are univariate, bivariate, and multivariate analyses?

These three analytical tiers are categorized by the number of variables examined concurrently:

• **Univariate Analysis:** Analyzes a **single variable** in isolation to inspect its frequency, central tendency (mean, median, mode), dispersion (variance, IQR), and distribution shape. Examples: Histograms, KDE plots, boxplots, countplots.
• **Bivariate Analysis:** Explores the relationship between **two variables** to detect correlations, group differences, or dependency patterns. Examples: Scatterplots, bivariate bar charts, cross-tabulations.
• **Multivariate Analysis:** Analyzes **three or more variables** concurrently to understand complex interaction effects, conditional relationships, and high-dimensional clustering. Examples: Heatmaps, pairplots, FacetGrids, 3D scatterplots, PCA biplots.

Question 6: Difference between heatmap and pairplot?

• **Heatmap (sns.heatmap):** Displays a 2D grid of aggregated statistical summaries (such as an N × N Pearson correlation matrix or confusion matrix) represented by colored tiles. It is computationally lightweight, summarizes macroscopic linear associations, but discards row-level granularity.
• **Pairplot (sns.pairplot):** Renders an N × N panel of individual subplots displaying actual raw data observations (scatterplots off-diagonal and univariate KDEs/histograms on diagonal). It allows segmenting data points with a categorical hue label, detects non-linear clusters, but is computationally heavier on large datasets.

Question 7: How do you summarize your insights?

An effective analytical summary bridges data engineering and executive business decision-making using the **BLUF (Bottom Line Up Front)** framework:

1. **Executive Headline:** State the overarching conclusion in 1-2 direct sentences.
2. **Quantify with Evidence:** Always substantiate findings with concrete percentages, metrics, and p-values (e.g., 'Female survival was 74.2% vs. 18.9% for males, $p < 0.001$').
3. **Visual Anchor:** Connect each finding to a named chart or figure with explicit callout captions.
4. **Actionable Business/Modeling Recommendations:** Translate findings into concrete engineering steps (e.g., log-transforming fares, imputing missing ages via class medians, creating family size features).

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6. Recommendations for Machine Learning Pipeline

Based on our exploratory findings, the following engineering steps are recommended before training classification models:

• **Feature Engineering:** Synthesize family_size = sibsp + parch + 1 and a categorical family tier (Solo, Small, Large). Create a title extractor from passenger names (Mr, Mrs, Miss, Master) which carries high predictive power for age and social standing.

• **Missing Value Imputation:** Impute age using median values grouped by pclass and sex. For embarked, impute mode ('S'). For deck, encode missingness as a binary indicator has_cabin.

• **Feature Transformation:** Apply np.log1p on fare to neutralize positive skewness and stabilize variance.